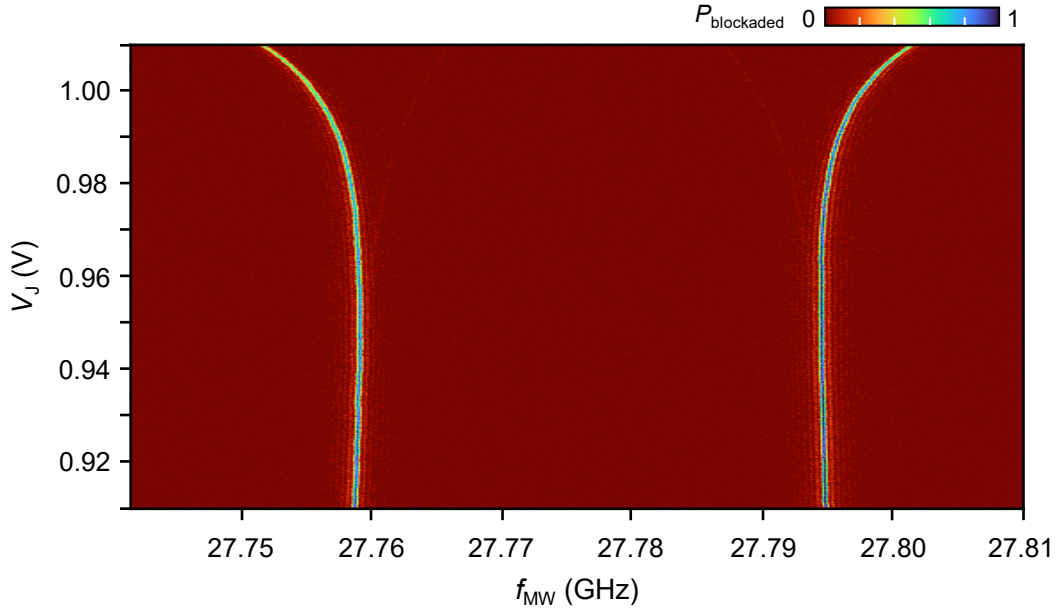
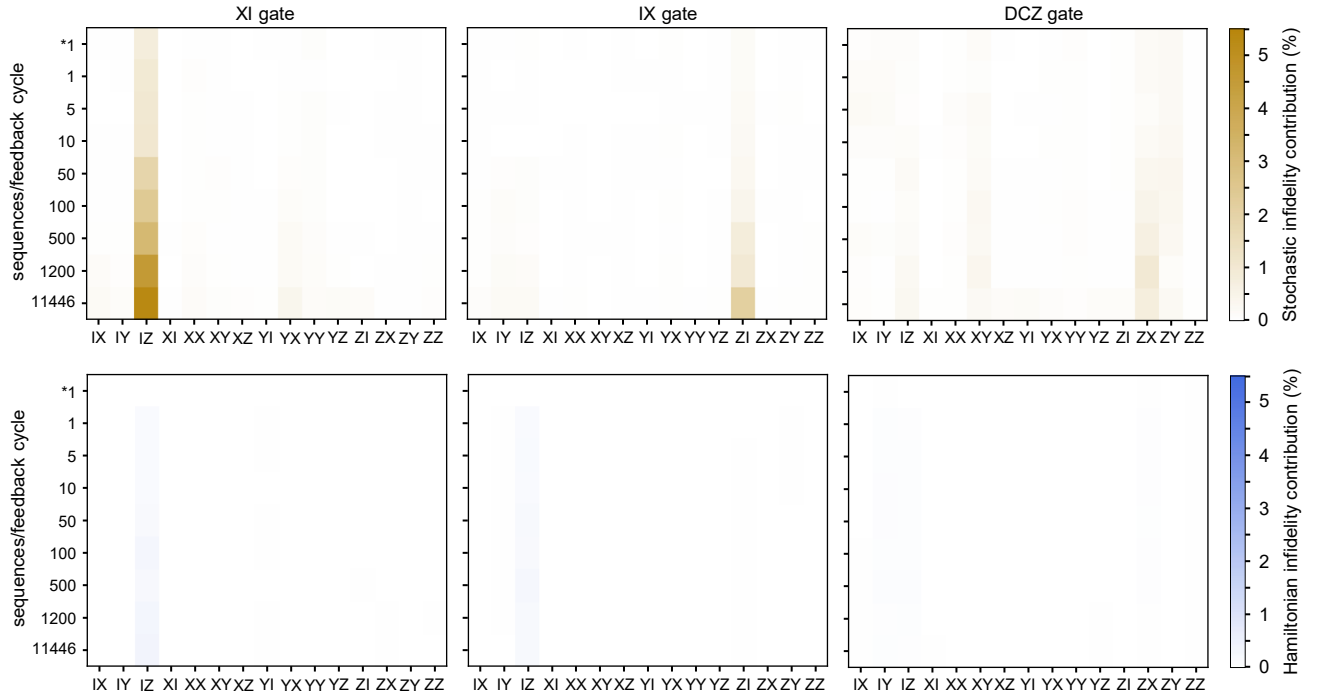


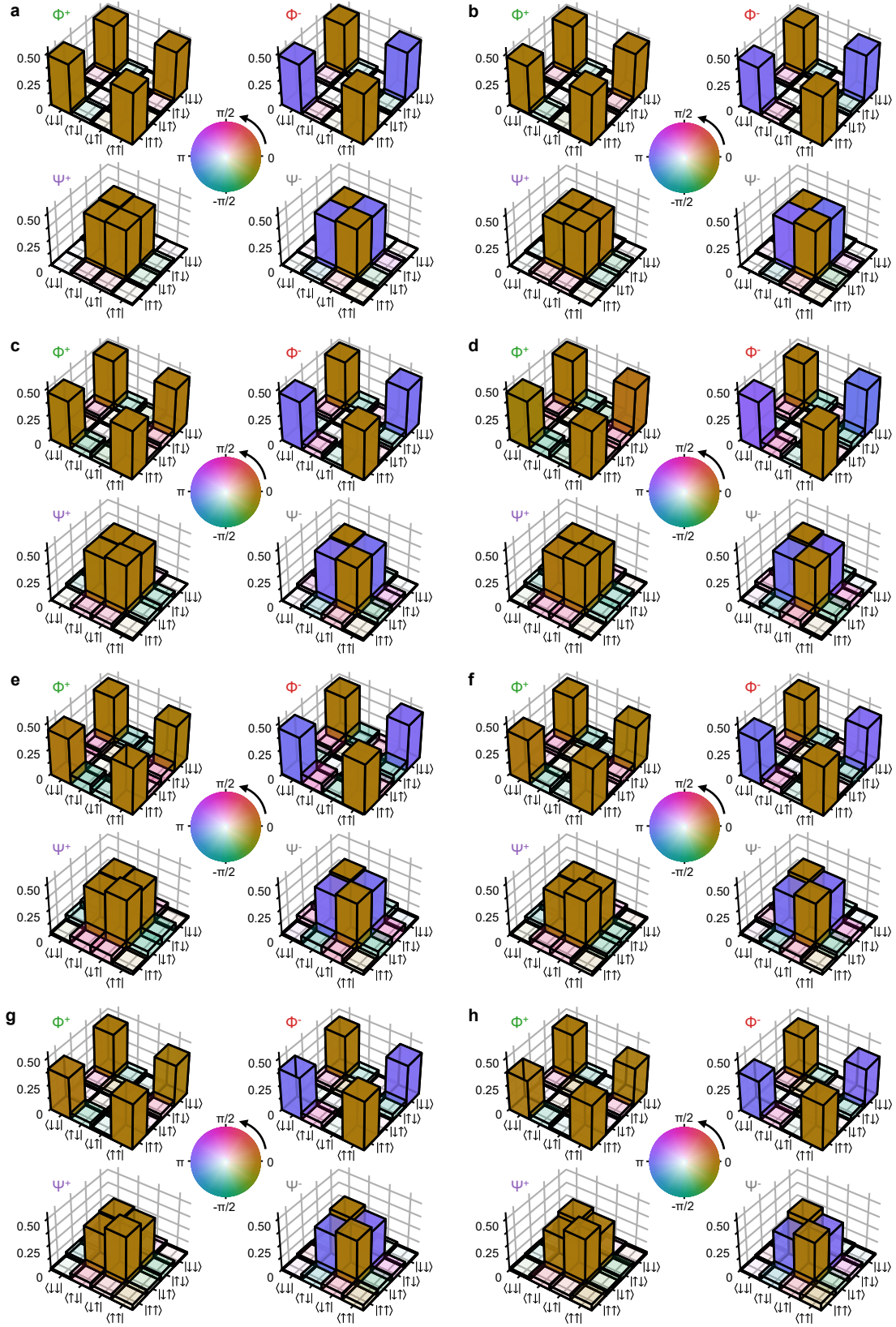
## SUPPLEMENTARY INFORMATION



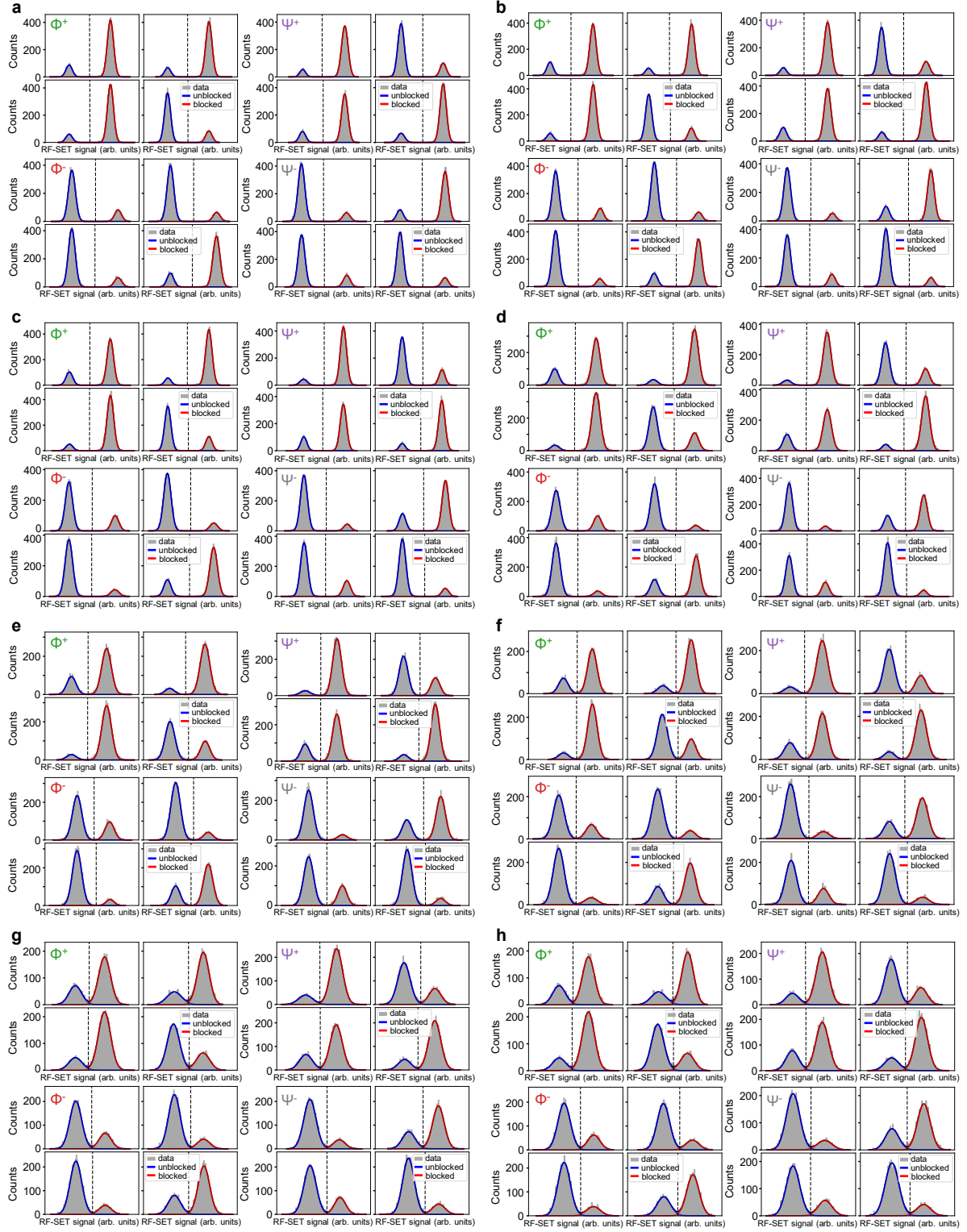
**Supplementary Fig. 1 | Odd state  $|\downarrow\uparrow\rangle$  initialization.** Probability of detecting a blockaded state,  $P_{\text{blockaded}}$ , after a microwave burst of fixed power and duration at different exchange gate voltage  $V_J$  when preparing a pure odd state  $|\downarrow\uparrow\rangle$ . The power and duration of the microwave burst are roughly calibrated to a single-qubit  $\pi$ -rotation.



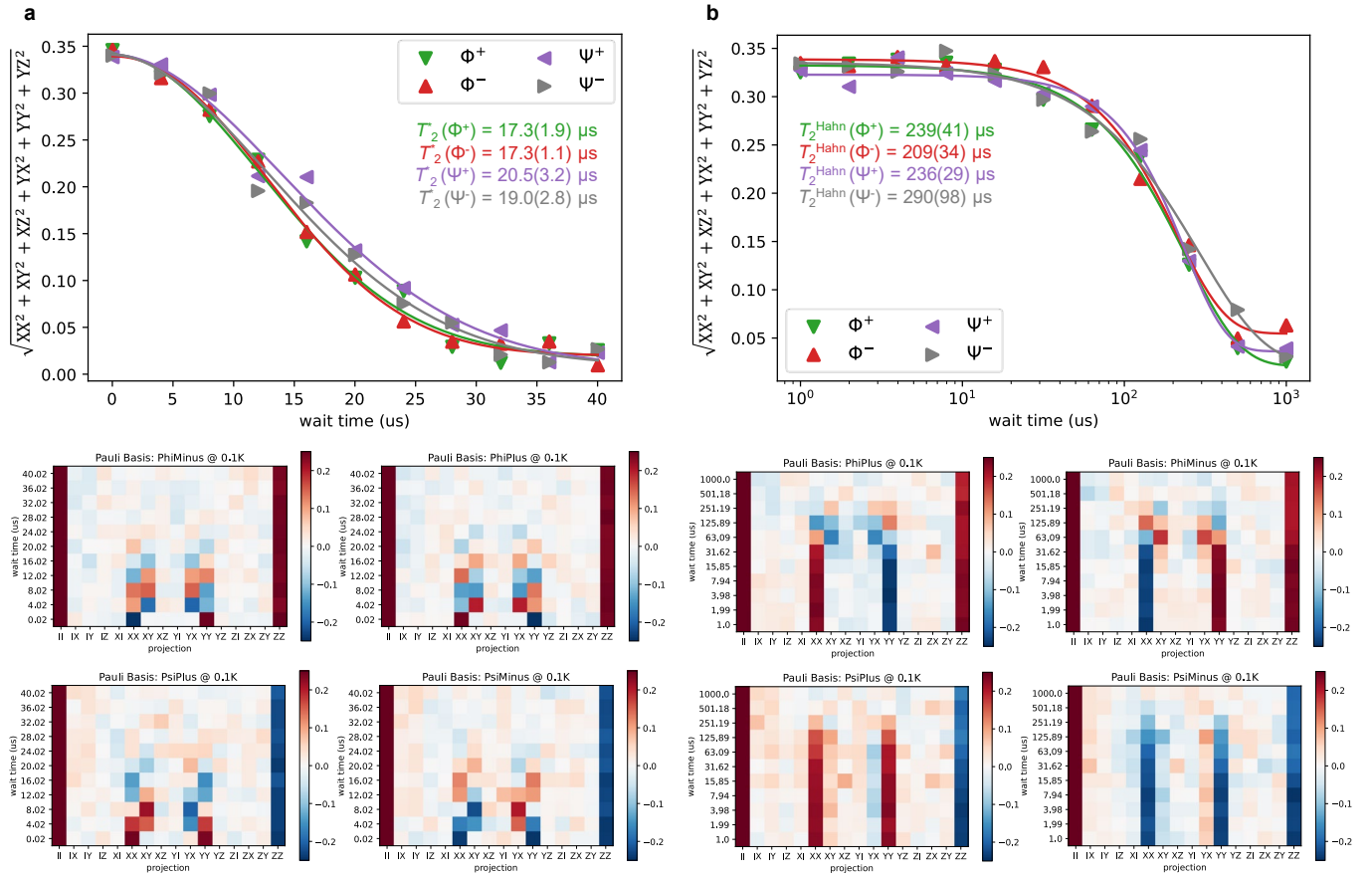
**Supplementary Fig. 2 | Breakdown of GST error channels.** Infidelity contributions of individual error channels using pyGSTi on Larmor frequency feedback rate measurement series including the additionally phase corrected experiment (\*) with fidelities of full 2-qubit gate set above 99%.



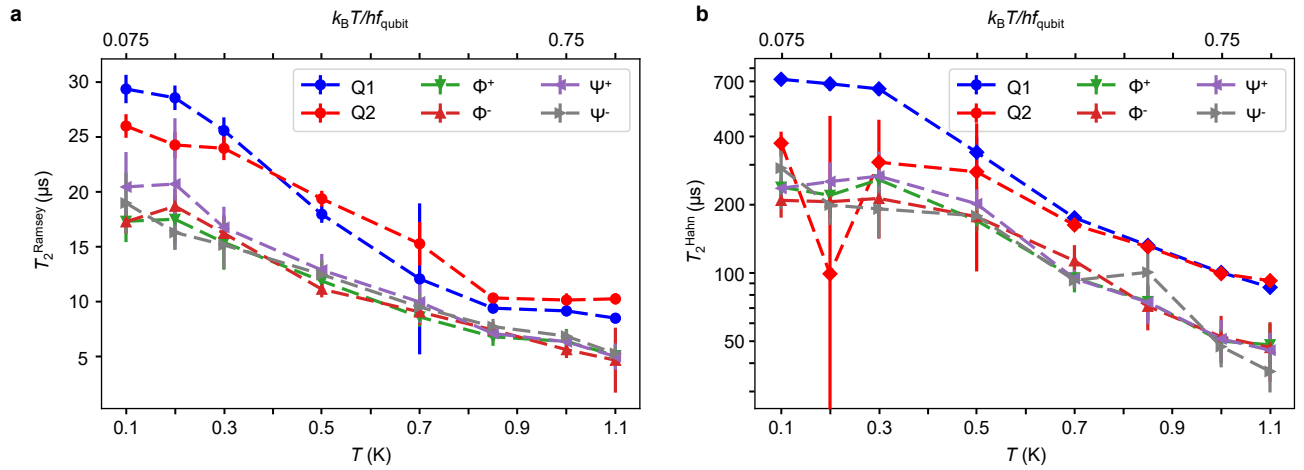
**Supplementary Fig. 3 | Bell state tomography as a function of temperature.** Density matrices of all four Bell states for temperatures **a**, 0.1 K, **b**, 0.2 K, **c**, 0.3 K, **d**, 0.5 K, **e**, 0.7 K, **f**, 0.85 K, **g**, 1.0 K, and **h**, 1.1 K extracted from quantum state tomography.



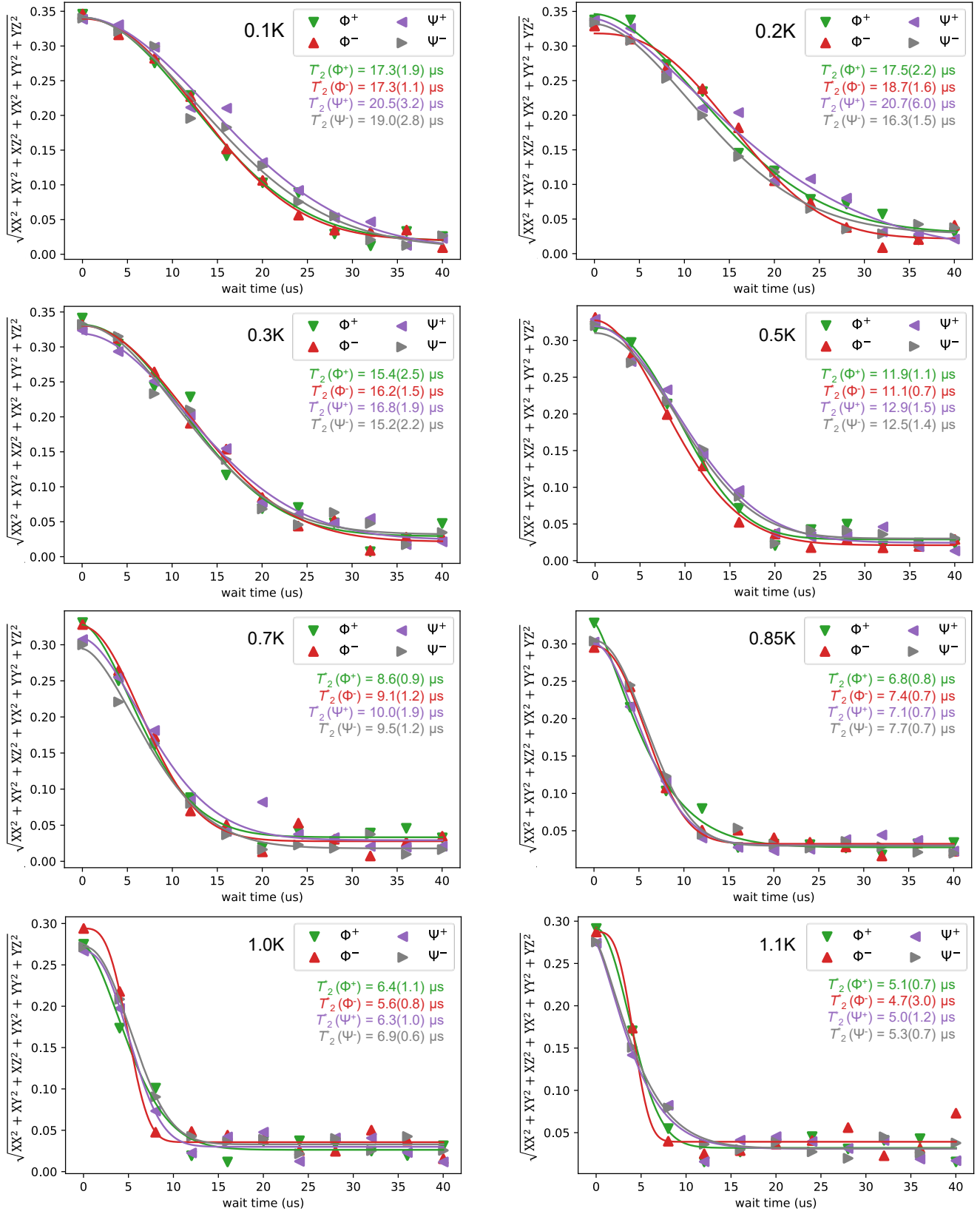
**Supplementary Fig. 4 | RF-SET signal histograms as a function of temperature.** RF-SET signal histograms of all four Bell states for temperatures **a**, 0.1 K, **b**, 0.2 K, **c**, 0.3 K, **d**, 0.5 K, **e**, 0.7 K, **f**, 0.85 K, **g**, 1.0 K, and **h**, 1.1 K obtained during the direct parity Bell experiment.



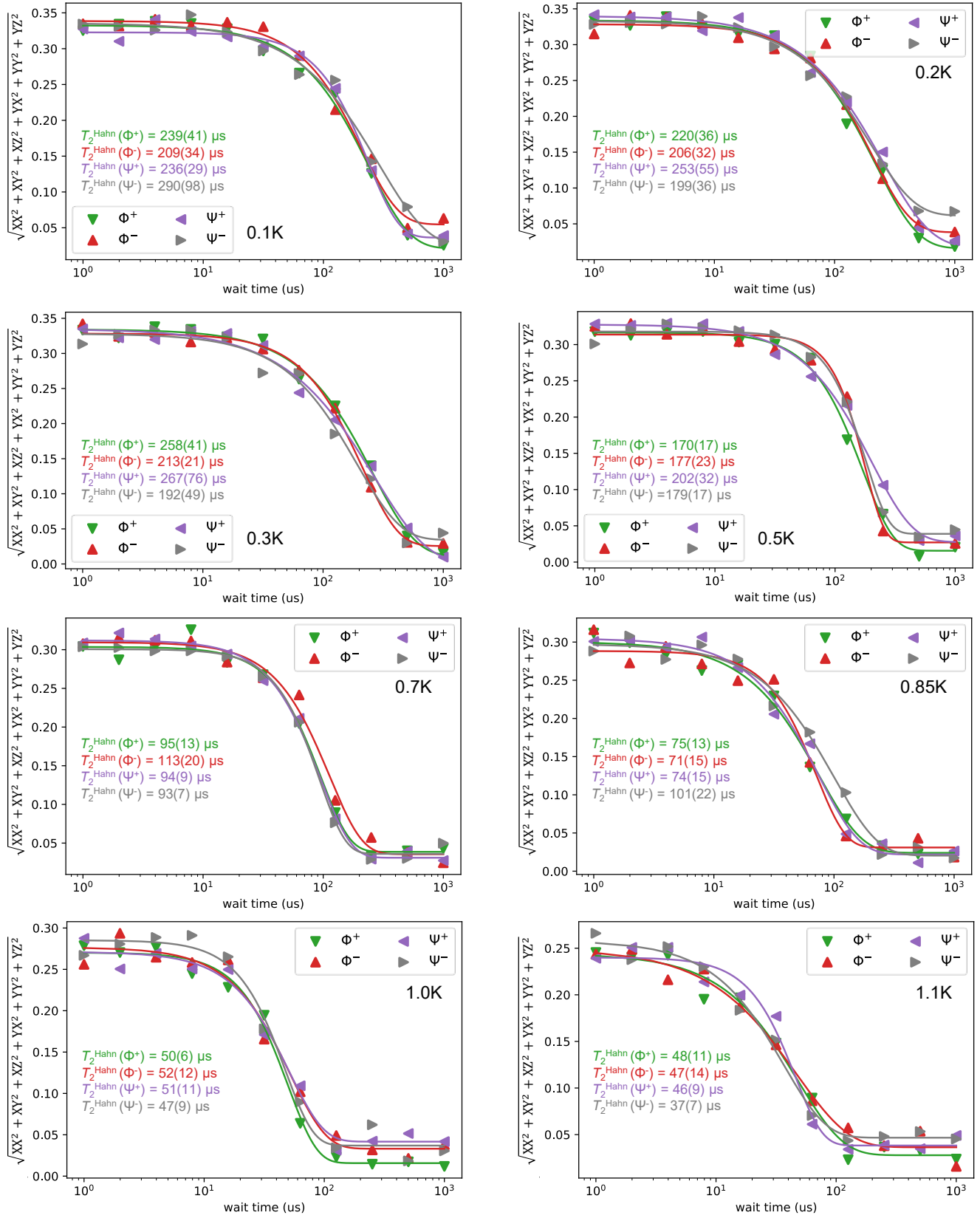
**Supplementary Fig. 5 | Pauli projections of Bell state lifetime experiments.** Pauli projections of all four Bell states as a function of wait time for **a**, Ramsey and **b**, Hahn experiment. The lower panels show the individual Pauli projections. In the upper panels we fit the squared sum of the contributing Paul projections to an exponential decay to extract all four Bell state lifetimes. Error bars represent the 95 % confidence level.



**Supplementary Fig. 6 | Coherence times as a function of temperature.** **a**, Ramsey and **b**, Hahn echo coherence times of the single qubits and all four Bell states as a function of temperature. Error bars represent the 95 % confidence level.

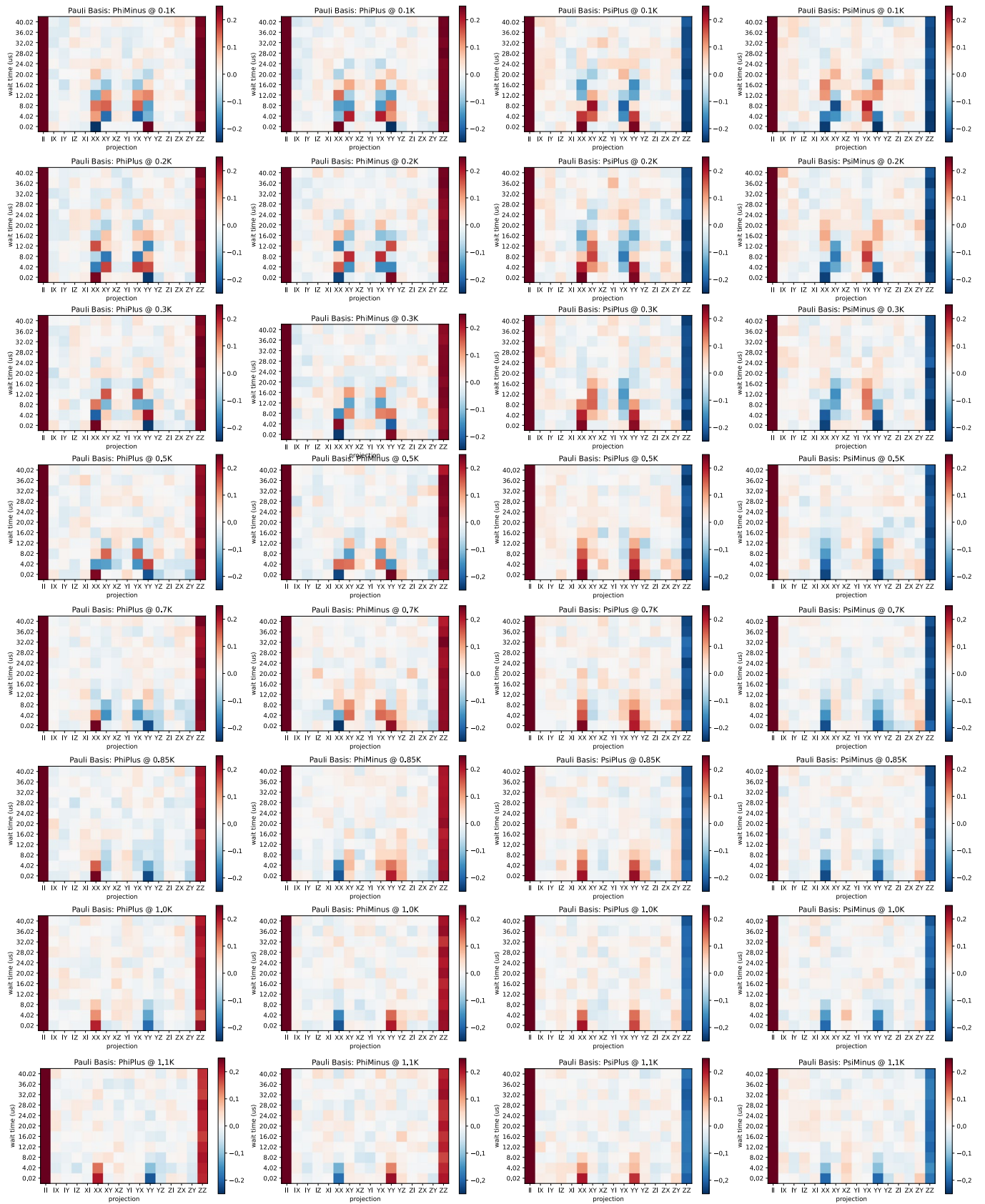


**Supplementary Fig. 7 | Ramsey Pauli projections as a function of temperature – squared sum.** The solid lines are exponential decays fitted to the data to extract the coherence times. Error bars represent the 95 % confidence level.

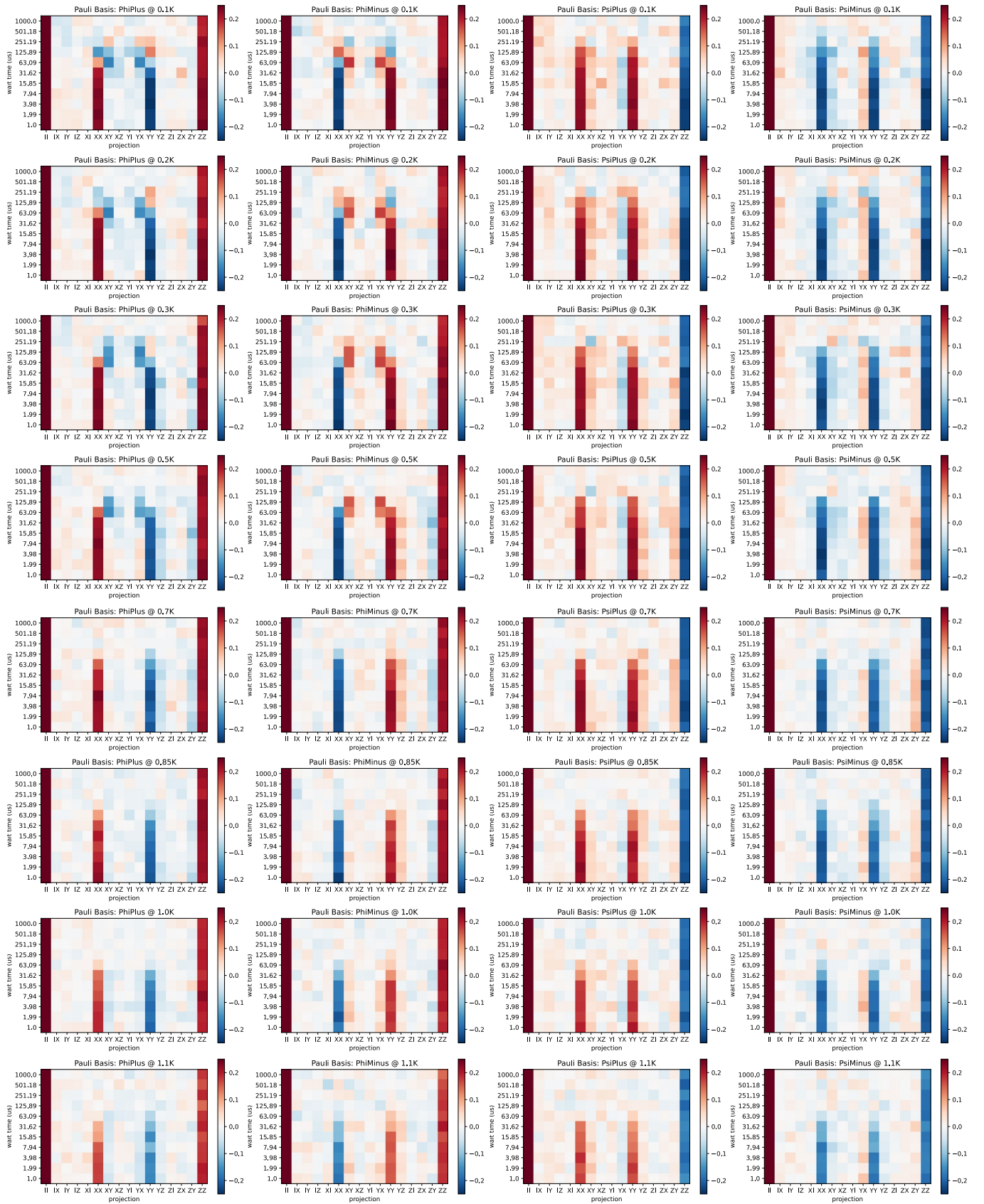


**Supplementary Fig. 8 | Hahn echo Pauli projections as a function of temperature – squared sum.** The solid lines are exponential decays fitted to the data to extract the coherence times. Error bars represent the 95 % confidence level.





Supplementary Fig. 9 | Ramsey Pauli projections as a function of temperature – individual projections.



Supplementary Fig. 10 | Hahn echo Pauli projections as a function of temperature – individual projections.